

Chemical Bonding

Page No.

Date

Chemical Bond

Strong Bonds
(≈ 200 KJ)

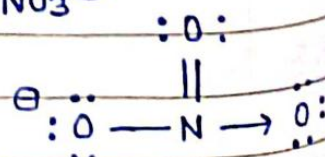
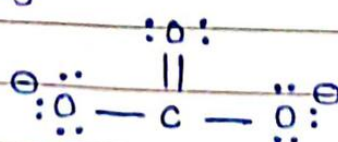
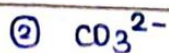
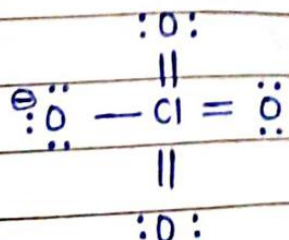
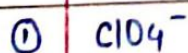
Weak Bonds
(2 to 40 KJ)

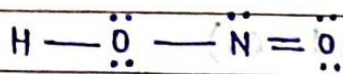
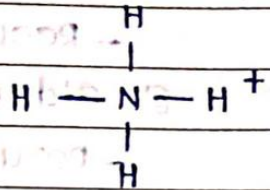
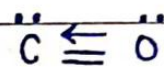
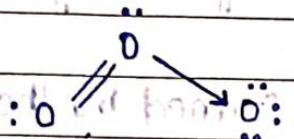
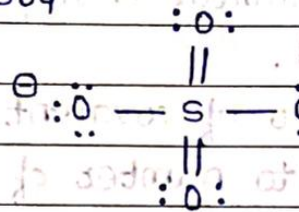
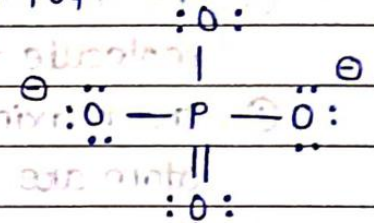
①	Covalent Bond	Vanderwaals	Hydrogen
②	Ionic Bond	Force	Bond
③	Co-ordinate Bond	(2 to 10 KJ)	(10 to 40 KJ)
④	Metallic bond		

- ① Covalent Bond - mutual sharing of e^-
- ② Ionic Bond - complete transfer of e^-
- ③ Co-ordinate Bond - shared e^- pair is contributed by only one atom.
- ④ Metallic Bond - force between kernel and Free e^- .

Lewis Dot structure:

- ① Identify the central atom and surrounding atom.
- ② Assign -ve charge to more electronegative atom and +ve charge.
- ③ O atom form two bonds but O^- form one bond.
- ④ Try to complete octet and duplet of all atoms.
- ⑤ But if central atom contains vacant d-orbital such as P, S, Cl etc then octet can expand.



④ HNO_2 ⑤ NH_4^+ ⑥ CO ⑦ O_3 ⑧ SO_4^{2-} ⑨ PO_4^{3-} 

- Formal charge: $\text{Valence } e^- - \text{one pair } e^- - \text{no. of surrounding Bonds}$

$$\text{Formal charge} = \text{Valence } e^- - \text{one pair } e^- - \text{no. of surrounding Bonds}$$

- Bond order From Resonance:

$$\text{B.O} = \frac{\text{Total no. of Bonds}}{\text{Total no. of positions}}$$

$$\text{Total no. of positions} = (3) (2) (2)$$

- Average Formal charge on surrounding atom:

$$= \frac{\text{Net charge on species}}{\text{no. of surr. atoms}}$$

- Valence Bond Theory (VBT):

① A covalent Bond is a result of overlap of atomic orbitals of two atoms.

② The atomic orbitals which overlap must contain unpaired electron with opposite spins. Also they must possess either same or nearly same energies.

③ Strength of covalent bond $\propto$ extent of overlap.

④ Two types of overlap — ① Head to Head (axial)

— Result in σ bond.



② side wise (lateral)

— Results in π bond.

⑤ The direction of orbital overlap decides geometry of molecule formed.

⑥ The maximum no. of covalent bonds formed by the particular atom are equal to number of the unpaired electron in its ground state.

ex: C in G.S — $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^0$

— carbon can form max. 2 bonds.

• Types of overlap :

① s-s overlap — H_2 formation — Always form σ bond.

② p-p overlap — $\text{Cl}_2, \text{F}_2, \text{O}_2$ — can form σ and π bond.
(σ) (σ) (σ and π)

③ s-p overlap — HCl Formation — always form σ bond.

HF Formation

• strength order :

$1s-1s > 1s-2p > 2p-2p > 2s-2p > 2s-2s > 3p-3p > 3s-3p > 3s-3s$

• Concept of Hybridisation (modified VBT)

— The mixing and recasting of atomic orbitals of same atom to form new orbitals called hybrid orbitals which are identical in all respect.

• Covalent Bond is a result of any of the following overlap:

- ① Hybrid orbital of one atom and atomic orbital of another atom.
ex: CH₄ Formation.
- ② Hybrid orbital of one atom and hybrid orbital of another atom.
ex: Graphite.

• Types of hybridisation:

- ① $sp - (2) -$ Linear
- ② $sp^2 - (3) -$ Trigonal planar
- ③ $sp^3 - (4) -$ Tetrahedral
- ④ $sp^3d - (5) -$ Trigonal Bipyramidal
- ⑤ $sp^3d^2 - (6) -$ Octahedral
- ⑥ $sp^3d^3 - (7) -$ Pentagonal Bipyramidal
- ⑦ $dsp^2 - (4) -$ Square planar
- ⑧ $d^3s - (4) -$ Tetrahedral
- ⑨ $dsp^3 - (5) -$ Trigonal Bipyramidal
- ⑩ $d^2sp^3 - (6) -$ Octahedral
- ⑪ $d^3sp^3 - (7) -$ Pentagonal Bipyramidal
- ⑫ $sp^2d - (4) -$ Square planar

• Method of Finding hybridisation:

$X \rightsquigarrow$ Indicate valence e^- of central atom.

- If surrounding atom is monovalent (H, F, Cl, Br, I, etc) then

allot 1 'X' form 1 atom.

- If surrounding atom is divalent (O, S) then allot 2 'X'

Form 1 atom.

- $H = \text{bond} + \text{no. of lone pair}$

If $H = 2 \rightarrow sp$ $H = 5 \rightarrow sp^3d$

$H = 3 \rightarrow sp^2$ $H = 6 \rightarrow sp^3d^2$

$H = 4 \rightarrow sp^3$ $H = 7 \rightarrow sp^3d^3$

• valence shell electron pair repulsion (VSEPR Theory):

- ① If central atom is surrounded by only bond pairs and no lone pairs then geometry and shape both are same. ex. CH_4 .
- ② If central atom is surrounded by both bond pairs and lone pairs then geometry and shape both are different.
- ③ The order of repulsion of various types of e^- pairs is
 $lp-lp > lp-bp > bp-bp$.
- ④ lone pair of central atom reduces bond angle from expected value.
 Decrease in bond angle from expected value $\propto$ no. of lone pairs on central atom.

Code number	Hybridisation	Geometry	Shape
220	sp	Linear	Linear ($CO_2, BeCl_2$)
330	sp^2	Trigonal planar	Trigonal planar (BF_3)
321	sp^2	"	V-shape ($SnCl_2$)
440	sp^3	Tetrahedral	Tetrahedral (CH_4)
431	sp^3	"	Pyramidal (NH_3)
422	sp^3	"	V-shape (H_2O)
550	sp^3d	TBP	TBP (PCl_5)
541	sp^3d	"	See-saw (SF_4)
532	sp^3d	"	T-shape (ClF_3)
523	sp^3d	"	Linear (XeF_2)
660	sp^3d^2	Octahedral	Octahedral (SF_6)
651	sp^3d^2	"	Square pyramidal (BrF_5)

642	sp^3d^2	square planar (XeF_4)
770	sp^3d^3	PBP (IF_7)
761	sp^3d^3	distorted octahedral
752	sp^3d^3	Pentagonal planar.

Bond parameters (measurable properties of covalent bond):

① Bond angle - angle between two adjacent covalent bonds.

Hybⁿ expected B.A. $\propto$ s% character (H is diff)

sp	180°	sp	sp ²	sp ³
sp ²	120°	50% s	33.33% s	25% s
sp ³	109° 28'			
sp ³ d	90°, 120°			
sp ³ d ²	90°, 180°			
sp ³ d ³	90°, 72°			
dsp ²	90°			

② No. of lone pairs

I. Same - Hybridisation, surrounding atom

Different - central atom, no. of lone pairs then

$$B.A \propto \frac{1}{\text{no. of lone pairs}} \quad \text{ex: } CH_4 > NH_3 > H_2O$$

II. Same - Hybridisation, surrounding atom, lone pair.

Different - central atom

$$B.A \propto EN \text{ of central atom} < NH_3 > PH_3 > AsH_3 > SbH_3$$

$$H_2O > H_2S > H_2Se > H_2Te$$

III. same $\rightarrow$ Hybridisation, central atom, lone pair

Different $\rightarrow$ Surrounding atom



level $B \cdot A \propto \frac{1}{EN}$

level $B \cdot A \propto \frac{1}{EN}$ of surr. atom.

- Bond order: The number of bonds between two atoms. Calculated by using any of the following:

① MOT — Total $e^- \leq 20$, species is diatomic.

② Resonance $\rightarrow B.O = \frac{\text{total no. of Bonds}}{\text{total no. of positions}}$

③ Use structure.

- Bond energy: Energy required to break 1 mole of particular bonds.

① Bond energy $\propto$ Bond multiplicity

② Bond energy $\propto$ %s character.

③ Bond energy $\propto \frac{1}{\text{atomic size}}$

④ Bond energy $\propto$ Bond polarity

⑤ Bond energy $\propto$ no. of lone pairs (use this factor when there is single bond betn 2 identi. bond)

- Big exception:

Bond energy in halogens

Expected order: $F-F > Cl-Cl > Br-Br > I-I$

Actual order: $Cl-Cl > Br-Br > F-F > I-I$

- **Bond length:** The distance between the centres of nuclei of two closely bonded atoms.

① $B.L \propto \frac{1}{\text{Bond multiplicity}}$

② $B.L \propto \text{Atomic size}$

③ $B.L \propto \frac{1}{\% \text{ s-character}}$

④ $B.L \propto \frac{1}{\text{Bond order}}$

Bond order

ex:

①

O₂

O₃

H₂O₂

←

B.O

Bond length →

② CO

CO₂

CO₃²⁻

Bond length

- **Dipole moment:** Measure of polarity of bond.

$$\mu = q \times r$$

Units: esu·cm, C·m, Debye.

$$1D = 10^{-18} \text{ esu} \cdot \text{cm}$$

$$1D = 3.33 \times 10^{-30} \text{ C} \cdot \text{m}$$

$$\mu_R = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos \theta}$$

$$\% \text{ ionic character} = \frac{\mu_{\text{observed}}}{\mu_{\text{theory}}} \times 100$$

μ_{theory}

Polar molecule = $\mu_R \neq 0$ (permanent dipole moment)

Non-polar molecule = $\mu_R = 0$

• Molecular orbital theory: (Applicable only for diatomic species having total $e^- \leq 20$)

① During molecule or ion formation through covalent bonding, Atomic orbitals combine together to form new orbitals called molecular orbitals.

② The number of molecular orbitals formed are exactly equal to number of Atomic orbitals combined.

③ Two types of Molecular orbitals formed:

a) Bonding MO. (denoted by σ and π)

b) Anti-Bonding MO's (denoted by σ^* and π^*)

AO's combined	BMO's combined	ABMO's formed
$ns + ns$	σns	$\sigma^* ns$
$np_z + np_z$	σnp_z	$\sigma^* np_z$
$np_x + np_x$	πnp_x	$\pi^* np_x$
$np_y + np_y$	πnp_y	$\pi^* np_y$

• Filling of electrons in MO's:

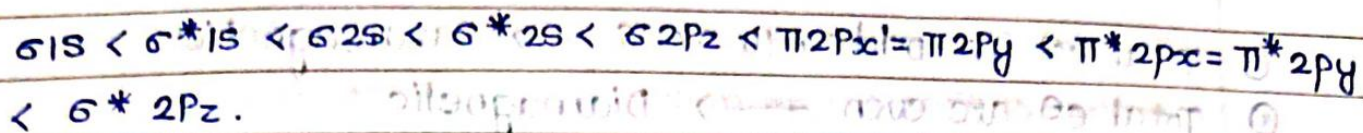
① Configuration I: when total $e^- \leq 14$.

ex: N_2 , N_2^+ , B_2 , C_2 etc (2s-2p mixing occurs)

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x = \pi 2p_y < \sigma 2p_z < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$

BABA BBB AAA — Trick

② Configuration II: when total $e^- = 15$ to 20, ex: O_2 , O_2^- , F_2 , Ne_2 etc (2s-2p mixing do not occurs)



BABA BBB AAA

Applications of MOT:

- ① Bond order = $B.O = \frac{1}{2} (N_b - N_a)$ If $B.O = 0$ species do not exist
 $B.O \neq 0$ species exist

Total e^- B.O

1

0.5

Total e^- B.O

9

0.5

2

1.0

3

0.5

4

0

5

0.5

6

1.0

7

0.5

8

0

10

1.0

12

2.0

13

2.5

14

3.0

15

2.5

16

2.0

17

1.5

18

1.0

19

0.5

20

0

- ② Bond length $\propto \frac{1}{\text{Bond order}}$

- ③ Bond energy $\propto$ Bond strength
 Bond strength $\propto$ Bond order.

Thermal stability

- ④ Prediction of magnetic nature:

- ① Total e^- are odd or 10 or 16 $\rightarrow$ Paramagnetic
② Total e^- are even $\rightarrow$ Diamagnetic.

• Ionic Bond :

(The Bond which is formed by complete transfer of e^- from one atom to another atom).

- ① $\Delta EN > 1.6$
- ② Low IE of element forming cation.
- ③ High electron gain enthalpy with -ve sign of element forming anion.
- ④ High lattice enthalpy.
- ⑤ Overall decrease in energy of system.

• Imp. note :

- ① The Bond Formed between metal and non metal is generally ionic.
- ② The Bond Formed between two complex ions or in between simple ion and complex ion is also ionic.
- ③ There are some substance which contains ionic, covalent as well as co-ordinate bond.

ex: NH_4Cl

$[Ni(H_2O)_6]SO_4$

- ④ Electrovalency : no. of electron lost or gained by one atom.

- Lattice energy or lattice enthalpy (Δ_{LH})
Amount of energy released when 1 mole of ionic compound is formed from constituent gaseous cation and gaseous anion. $\Delta_{LH} = -ve$.

or Amount of energy required to break 1 mole of ionic compound into its constituent gaseous cation and gaseous anion.

$$\Delta L H = +ve.$$

$$X_{80} > X_{70}, X_{100} > X_{90} : X_{90}$$

• V. Imp:

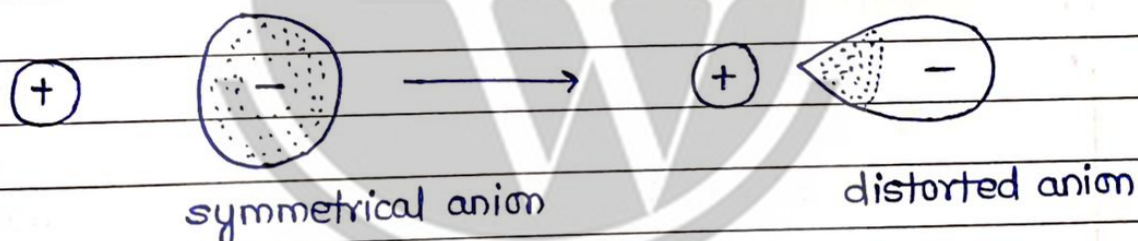
① stability of ionic compound $\propto \Delta L H$.

② solubility of ionic compound in $H_2O \propto$ lattice enthalpy

$$\Delta L H \propto \frac{q_c \cdot q_a}{r_c + r_a}$$

$$\Delta L H \propto \frac{q_c \cdot q_a}{r_c + r_a}$$

• covalent nature in ionic bond: (concept of polarisation)



For cation $\rightarrow$ polarising power.

For anion $\rightarrow$ polarisability

covalent character in ionic substance $\propto$ extent of polarisation

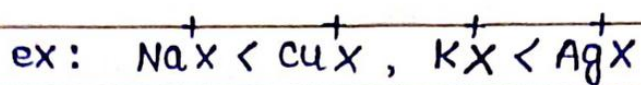
• Fajan's rules:

① Polarisation $\propto \frac{\text{size of anion}}{\text{size of cation}}$

② Polarisation $\propto q_c \text{ and } q_a$.

SHILA
small cation $\swarrow$ $\downarrow$ $\searrow$ large anion
High charges

③ Polaxising power is exceptionally more if cation possess pseudo inert gas configuration ($ns^2np^6nd^{10}$)



• V. Imp:

Melting point, thermal stability,
ionic nature, solubility in H_2O

Polarisation

(ability to distort electron cloud of anion)



distorted anion

symmetrical anion

covalent character in ionic substance & extent of polarisation

Polarising power:

Size of cation

Size of anion

Charge of cation

Charge of anion

Polarisation & size of anion

Size of cation

Polarisation & charge of cation